$$\frac{\Gamma, A \vdash B}{\Gamma \vdash A \Rightarrow B}\ \textsc{Imp-I} \qquad \frac{\Gamma \vdash A \Rightarrow B \qquad \Gamma \vdash A}{\Gamma \vdash B}\ \textsc{Imp-E}$$

$$\frac{\Gamma \vdash A \qquad \Gamma \vdash B}{\Gamma \vdash A \wedge B}\ \textsc{And-I} \qquad \frac{\Gamma \vdash A \wedge B}{\Gamma \vdash A}\ \textsc{And-EL} \qquad \frac{\Gamma \vdash A \wedge B}{\Gamma \vdash B}\ \textsc{And-ER}$$

$$\frac{\Gamma \vdash A}{\Gamma \vdash A \vee B}\ \textsc{Or-IL} \qquad \frac{\Gamma \vdash B}{\Gamma \vdash A \vee B}\ \textsc{Or-IR} \qquad \frac{\Gamma \vdash A \vee B \qquad \Gamma, A \vdash C \qquad \Gamma, B \vdash C}{\Gamma \vdash C}\ \textsc{Or-E}$$

$$\frac{\Gamma \vdash A \qquad x \notin FV(\Gamma)}{\Gamma \vdash \forall x.A}\ \textsc{All-I} \qquad \frac{\Gamma \vdash \forall x.A}{\Gamma \vdash A[x \leftarrow t]}\ \textsc{All-E}$$

$$\frac{\Gamma \vdash A[x \leftarrow t]}{\Gamma \vdash \exists x.A}\ \textsc{Ex-I} \qquad \frac{\Gamma \vdash \exists x.A \qquad \Gamma, A \vdash C \qquad x \notin FV(\Gamma, C)}{\Gamma \vdash C}\ \textsc{Ex-E}$$

$$\frac{}{\Gamma, A, \Delta \vdash A}\ \textsc{Ax} \qquad \frac{\Gamma \vdash A \qquad A \equiv_\beta B}{\Gamma \vdash B}\ \textsc{Conv}$$

$$\frac{\Gamma \vdash \bot}{\Gamma \vdash A}\ \textsc{Bot-E} \qquad \frac{}{\Gamma \vdash A \vee \neg A}\ \textsc{PEM}$$